

6 a Since both lamp are connected in parallel, they have the same p.d. of 2.7 V. **M1**

Total current = current of P + current of Q

$$= 0.15 + 0.090$$

$$= 0.24 \text{ A}$$

A1

b $E = V_R + V_P \text{ (or } V_Q)$

$$4.5 = V_R + 2.7$$

C1

$$V_R = 1.8 \text{ V}$$

$$I_R R = 1.8$$

$$0.24 \times R = 1.8$$

$$R = 7.5 \Omega$$

A1

c $R = \frac{\rho l}{A}$

$$\frac{R_P}{R_Q} = \frac{I_P}{I_Q} \times \frac{A_Q}{A_P} = \frac{I_P}{I_Q} \times \frac{d_Q^2}{d_P^2} \quad \text{C1}$$

$$\frac{R_P}{R_Q} = \frac{2.7 / 0.15}{2.7 / 0.09} = 0.60 \quad \text{C1}$$

$$0.60 = \frac{I_P}{I_Q} \times \frac{d_Q^2}{(2d_Q)^2}$$

$$\frac{I_P}{I_Q} = 2.4 \quad \text{A1}$$

- d** When filament wire of lamp Q break, the total resistance of the circuit increases and current in the circuit decreases. **B1**

This decreases the p.d. across the resistor and increase the p.d. across P. Hence the resistance of lamp P increases. **B1**

Or

When filament wire of lamp Q break, there is only one path for the current to flow through and the current flowing through P increases. **B1**

Hence, from Fig. 6.2, the increase in current in P is accompanied by a more than proportional increase in voltage across P. Hence the resistance of lamp P increases. **B1**